

Area Under a Curve Using Limits of Sums

Date_____ Period____

Evaluate each sum.

1) $\sum_{k=1}^n 48k$

2) $\sum_{k=1}^n 16k^2$

3) $\sum_{k=1}^n (36k + 6)$

4) $\sum_{k=1}^n (k + 2)$

Evaluate each limit.

5) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{12k}{n^2}$

6) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{2}{n} + \frac{k^2}{n^3} \right)$

7) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{6}{n} + \frac{16k^2}{n^3} \right)$

8) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{12}{n} + \frac{8k^2}{n^3} \right)$